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TabaPort: GIS and Geomatics Component

Locating Kampala's flood-prone areas, siting the first sensors and preparing the offline map for dispatch

Submitted to: Makerere Innovation and Incubation Center (MIIC), Technology Entrepreneurship Seed (TES) Programme
Project: TabaPort: Offline-first flood intelligence and dispatch for Kampala
Project lead: Shakiran Nannyombi
Component author: Mugisha Trevour Jeanclaude
Status: IEEE Response Quest Concept #5267, advanced to Phase 3 (video due 9 October 2026)
Date: 28 September 2026

Executive summary for the MIIC TES review

TabaPort is an early-stage hardware and software build: a culvert sensor feeds an offline hub that colours Kampala's roads green, yellow or red and ranks where a KCCA crew should go first, even when the mobile network is down. This report presents the **GIS and Geomatics component**, which answers one question for the whole product: **where**.

What the component delivers: a register of 17 documented flood-prone places across three valley systems; a flood susceptibility model in Google Earth Engine using open data (weights checked, consistency ratio 0.007); a shortlist of five first-sensor sites led by Mulago roundabout and Kalerwe; a survey method that turns each sensor reading into a true water elevation; and every map file the offline dispatcher laptop needs, prepared legally and loaded before the rain.

What it needs from TES: mentorship on hydrology and drainage validation, help opening a data conversation with KCCA, and access to survey-grade GNSS and levelling equipment for five sites (Section 11).

This report complements the TabaPort concept note led by Shakiran Nannyombi and does not replace it. Figures quoted from the team's submissions are reproduced as written; where the GIS work adds or refines a point, the report says so explicitly (Section 3).

1. TabaPort at a Glance

Kampala flash floods close Clock Tower, Bwaise, Kalerwe and other blackspots within minutes. Mobile networks congest or drop, and drainage crews, emergency teams, boda riders and commuters cannot see which drain is rising, which road is passable or where to send the next truck. The concept note records KCCA's 69 flood blackspots, a multi-year civil works programme of about UGX 768 billion through 2030/31, and UNDP's estimate of about US$50 million a year in flood damage. Construction moves water; it does not tell responders where the water is while channels are being built, or after waste and silt block them.

Layer	What it does (from the concept note)	GIS contribution
IoT node	ESP32 + waterproof ultrasonic measures culvert water level; moving-median filter ignores floating plastic; target under US$55	Chooses the culvert; surveys the sensor height so readings become elevations
Edge hub	Laptop now, rugged hub later; local map; no cloud required	Supplies the local map: road segments, base tiles, flood-prone areas
Dispatch	Roads green / yellow / red; silent node = unknown, not safe; ranked job list; routing avoids flooded segments	Per-segment ranking inputs; routable road network; measured high ground
Human report	Optional USSD / SMS from a boda rider; phone numbers	Place-name gazetteer that puts a report on the map

Layer	What it does (from the concept note)	GIS contribution
	stripped at the door	with a 100 m circle

TabaPort does not stop flooding. It reduces blind deployment and people driving into water. The GIS component makes sure the hub knows the ground before the storm arrives.

2. Where the GIS Component Fits in the Team

Team member	Responsibility	Team submission
Shakiran Nannyombi	Project lead: concept, IEEE entry, dispatch priority logic, KCCA relationship	TabaPort concept note; IEEE Response Quest #5267
Eric	Drainage structures and sensor placement	Drainage Sensor Placement for Urban Flood Monitoring (technical note)
Sharif	Software spine: offline map, data model, dispatch loop	Research Document: Stack and Data
Mugisha Trevour Jeanclaude	**GIS and Geomatics: locations, flood susceptibility, survey, map data**	**This report**

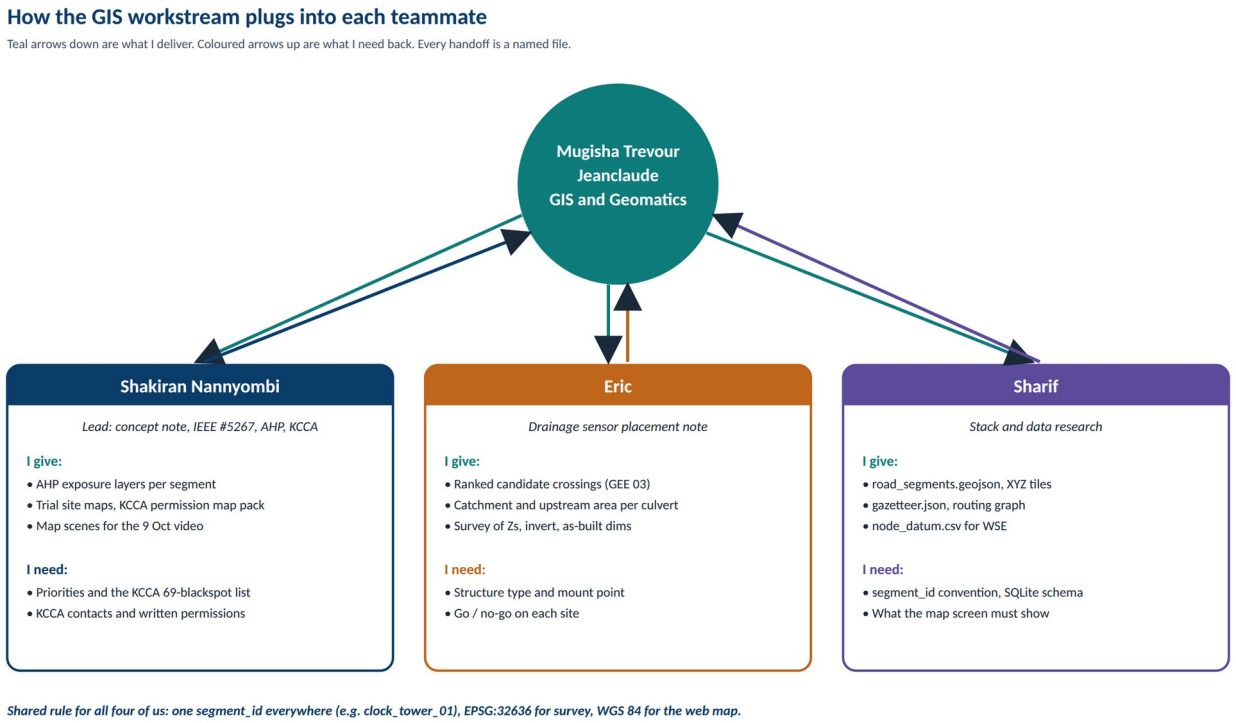


Figure 1. *How the GIS component connects to each teammate. Each arrow is a named file or decision, so responsibilities stay clear.*

2.1 Who I Am and What I Bring

I am a Bachelor of Science in Land Surveying and Geomatics student at Makerere University (expected completion August 2027), linked to the Makerere University GIS Centre. My working line is **Solutions Not Excuses**. The past work below feeds directly into TabaPort:

Past work	What it proves	Where it lands in TabaPort
FIG 2026 Congress, Cape Town: Geo-AI for malaria outbreak prediction (LSTM + Bayesian hierarchical model, DHIS2 + Sentinel-2); one of 10 FIG2026 Student Support Funding recipients	Satellite time series plus ground records, modelled to predict before the event	Satellite flood history (5.3); a later rise-rate forecast so upstream nodes warn downstream sites
LifeShield: pandemic preparedness command centre for Uganda (road-network spread model, Getis-Ord Gi* and Moran's I, DBSCAN)	Network distance, not straight lines; statistical proof that a cluster is real	Routing on the real road graph (8.2); hot spot confirmation after each storm (8.4)

Past work	What it proves	Where it lands in TabaPort
President, YouthMappers chapter at Makerere University	Organising volunteers to map in OpenStreetMap	Mapathons to fill missing lanes in Bwaise and Kalerwe (6.1)
Founder, IEEE Geoscience and Remote Sensing Society, Makerere University Student Branch Chapter	IEEE remote sensing community on campus	Direct fit with the IEEE Response Quest entry and peer review of the model
Secretary General, FIG Young Surveyors Network Uganda Chapter	A network of young surveyors and access to field instruments	GNSS and levelling support for site surveys (7.2)
Build with AI Makerere Hackathon, 27 June 2026 (GDG on Campus; co-organised by Shakiran Nannyombi); beVisioneers Mercedes-Benz Fellowship, Cohort 4	Shipping working prototypes fast, alongside this project's lead	Map layers delivered on time for the 9 October video

3. How This Report Stays Consistent with the Team's Submissions

MIIC will read four TabaPort documents. They must tell one story. The GIS component keeps them aligned in five practical ways:

1. **One source file per fact.** Locations live only in the flood location register and the road segments file. The other documents cite IDs (TP-01, clock_tower_01), never re-typed coordinates.
2. **Shared identifiers.** place_id (TP-nn), segment_id (Sharif's clock_tower_01 pattern) and node_id mean the same thing in every document, table and screen.
3. **Figures quoted as written, with source.** Numbers from the concept note (69 blackspots, 362 km, UGX 768 billion, US$50 million) are reproduced unchanged; outside figures carry a reference.
4. **A live alignment register.** Points where the documents describe the same thing differently are listed below with the GIS handling, and re-checked before every submission.
5. **Dated change log.** Every GIS file update is logged; when a teammate's design changes (for example the sensor type), the affected GIS files are updated within the week.

3.1 What Each Team Submission Needs from GIS

Team submission	What it establishes	What the GIS component supplies
Concept note (Shakiran Nannyombi)	Product in one sentence; 69 blackspots; silent node = unknown; ranked jobs; 30-day trial; no public-drain install without written KCCA permission	Located and ranked blackspots; routable segments; hospital and population layers; a surveyed trial site
IEEE Response Quest #5267	AHP dispatch on rise rate, critical infrastructure and population; A* routing with infinite penalty on submerged segments; gazetteer and 100 m radius for text reports	Per-segment AHP inputs; the road graph; the gazetteer itself
Eric: sensor placement note	Existing, stable structure at a documented blackspot; out of the debris zone; upstream and downstream levels for head difference; six-step pre-installation record	City-wide screen for candidate crossings; survey of positions and heights so upstream and downstream readings share one datum
Sharif: stack and data	Leaflet + local tiles + GeoJSON on localhost; SQLite with four tables; segment IDs; three screens; privacy by hashing	The GeoJSON segment file (the join key), a legally rendered tile set, gazetteer and router modules

3.2 Alignment Register

Topic	As written in team documents	How the GIS component handles it
Drainage network length	Concept note: 362 km	KCCA's published breakdown is 77.88 km primary, 134.46 km secondary and about 700 km tertiary (PreventionWeb); scope stated beside any figure
Detour routes	#5267: high-ground routes such as the Northern Bypass	High ground measured per segment (HAND); the Lubigi stretch of the Bypass is on KCCA's hotspot list

Topic	As written in team documents	How the GIS component handles it
Hub spatial engine	#5267: SpatiaLite on the hub. Stack research: SpatiaLite waits	All spatial work done before the rain; the hub reads plain GeoJSON
Public alerts	#5267: 2 km location-targeted texts. Stack research: cell broadcast is government-authorised	GIS prepares the 2 km alert area for KCCA's own channels; TabaPort does not broadcast
Generation 1 sensor	Concept note: JSN-SR04T ultrasonic. Eric: 80 GHz radar strongest overall	Survey method works for both; ultrasonic in a stilling tube for Generation 1 as Eric advises for budget builds
Node cost	#5267 p.4: $5 gauge. Elsewhere: under US$55	GIS materials use under US$55
Sampling interval	#5267: every 30 s. Stack research: 0.5 Hz	Rise rate defined over a fixed time window so ranking is stable either way
Offline base map	Stack research: pre-downloaded tiles	Tiles rendered locally from OpenStreetMap data (the OSM tile servers do not permit offline use)

4. Kampala's Flood Geography

Kampala is a city of hills and valley floors. Kololo Hill rises to 1,311 m and the Lake Victoria shore sits at about 1,135 m. The historic hills are separated by valleys holding sluggish streams and swamps, which once covered about 15% of the city (Kinawataka, Nakivubo, Lubigi, Kansanga, Kyetinda). Bwaise, Nateete and Katwe sit on valley floors at about 1,180 to 1,190 m.

KCCA drains the city through **eight primary channels**: Nalukolongo, Nakivubo, Lubigi, Kansanga, Mayanja Kaliddubi, Nalubaga, Kinawataka and Walufumbe. The Nakivubo channel runs about 9 km from Makerere Kivulu and Bat Valley through the CBD to the Nakivubo wetland, and was widened to a reinforced 12 m dual channel by March 2026. Rain falls in two wet seasons (February to June and August to December), with April the wettest month at about 169 mm.

The flood mechanism is geometric. Short, intense storms fall on sealed hillsides; runoff reaches the valley floors in minutes, where markets, junctions and settlements occupy former wetland and where culverts choked with plastic and silt act as dams. Recent events repeat the pattern: 26 to 28 March 2025 (GDACS, at least 7 deaths), 31 October 2025 along the Nakivubo, and July 2026 at Nalukolongo, Najjanankumbi, Lugogo and Kyambogo, after which Parliament gave KCCA seven days to explain.

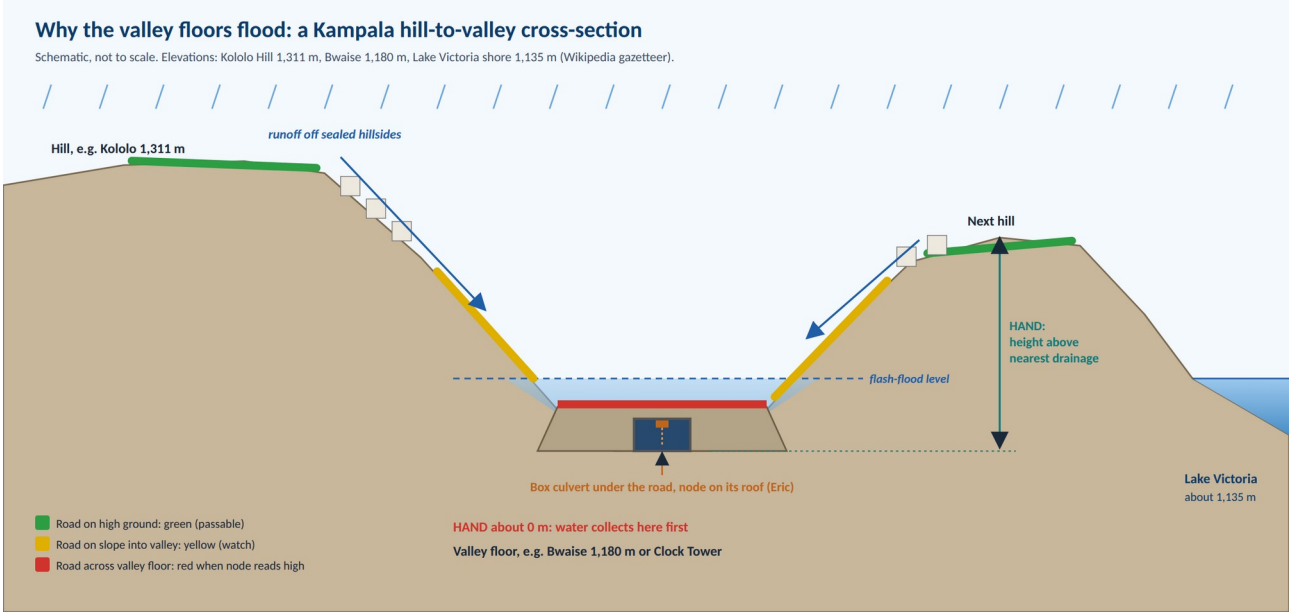


Figure 2. *Hill-to-valley cross-section. Roads on ridges stay green, slopes turn yellow, valley-floor crossings go red first. HAND is the height of a place above the drainage it would flood from.*

5. Locating the Flood-Prone Areas

The register combines every location named in the team's documents with KCCA's published hotspots and 2025 to 2026 reporting. Coordinates are either the Wikipedia gazetteer point for the neighbourhood (**W**) or an estimate by the author (**E**, expected within about 500 m). **All points are indicative until fixed by GNSS in the field**, which is the first field milestone (Section 10). Basins are provisional until confirmed from flow direction data.

ID	Place	Division	Basin (prov.)	Lat, Lon	Src	Why prone	Trial
TP-01	Clock Tower junction (Entebbe Rd / Queensway)	Central	Nakivubo	0.3095, 32.5785	E	Valley floor of the Nakivubo system below Nakasero; Kibuli and Mengo hills; several roads converge on one low crossing	High
TP-02	Bwaise	Kawempe	Lubigi (via Nsooba)	0.3500, 32.5611	W	Former wetland floor at about 1180 m; dense informal settlement on the drainage line	High
TP-03	Kalerwe market (Bombo Rd / Gayaza Rd)	Kawempe	Lubigi (via Nsooba)	0.3525, 32.5720	E	Market and junction on the Nsooba valley line; heavy solid waste load into drains	High
TP-04	Mulago roundabout	Kawempe	Lubigi (via Nsooba)	0.3395, 32.5745	E	Road dip at the foot of Mulago Hill beside the national referral hospital	High
TP-05	Kubiri roundabout (Bombo Rd)	Kawempe	Lubigi (via Nsooba)	0.3460, 32.5690	E	Low junction on Bombo Rd between Wandegeya and Bwaise	Medium
TP-06	Northern Bypass at Lubigi wetland	Kawempe / Rubaga	Lubigi	0.3470, 32.5500	E	Highway crosses the Lubigi wetland floor; not high ground on this stretch	High
TP-07	Nakivubo channel at St Balikuddembe market	Central	Nakivubo	0.3110, 32.5775	E	Encroached channel in the CBD valley; channel reconstructed to 12 m by Mar 2026	Low (reconstructed section)
TP-08	Lugogo Bypass	Nakawa / Central	Nakivubo (Lugogo channel)	0.3285, 32.5990	E	Low road along the Lugogo channel that feeds Nakivubo	Medium
TP-09	Mukwano Rd near Kibuli	Central / Makindye	Nakivubo	0.3070, 32.5930	E	Industrial area valley floor draining to Nakivubo wetland	Medium
TP-10	Nateete (Old Masaka Rd)	Rubaga	Nalukolongo	0.2983, 32.5350	W	Valley road at about 1190 m on the Nalukolongo headwaters	Medium
TP-11	Nalukolongo channel	Rubaga	Nalukolongo	0.2960, 32.5490	E	Primary channel reported as undersized for its catchment (Parliament Jul 2026)	High
TP-12	Ndeeba	Rubaga	Nalukolongo	0.2940, 32.5610	E	Low commercial strip on the Masaka Rd corridor	Medium
TP-13	Kinawataka	Nakawa	Kinawataka	0.3350, 32.6330	E	Kinawataka swamp floor; primary channel	Medium
TP-14	Kyambogo	Nakawa	Kinawataka	0.3480, 32.6290	E	Low ground below Kyambogo hill on the Kinawataka side	Medium
TP-15	Ntinda (factory close)	Nakawa	To confirm in GEE	0.3550, 32.6144	W	Valley between Ntinda and Naguru hills	Medium
TP-16	Kansanga / Kasanga-Ggaba	Makindye	Kansanga	0.2872, 32.6078	W	Natural low-lying terrain on the Kansanga swamp line	Medium
TP-17	Port Bell	Nakawa	Lake Victoria shore	0.2890, 32.6530	E	Near lake level at about 1135 m; backwater from the lake	Low

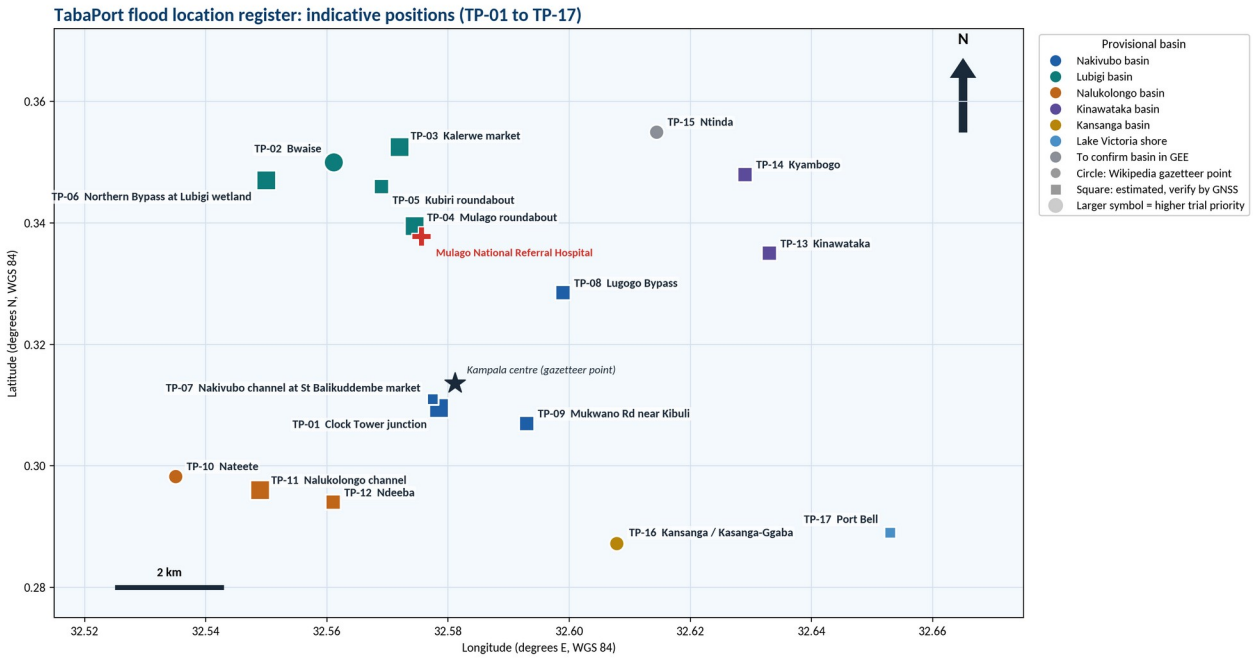


Figure 3. *Indicative positions of the 17 register points by provisional basin. Larger symbols mark higher trial priority. Mulago National Referral Hospital is shown because it drives the critical-infrastructure weight at TP-04.*

5.1 Key Findings

- **Three valley systems.** The north (Bwaise, Kalerwe, Kubiri, Mulago roundabout, Northern Bypass) drains towards Lubigi; the centre (Clock Tower, Nakivubo, Mukwano, Lugogo) to the Nakivubo; the south-west (Nateete, Nalukolongo, Ndeeba) to the Nalukolongo.
- **A natural sensor chain.** Mulago roundabout, Kubiri, Kalerwe and Bwaise appear to sit on one drainage line. If flow-direction analysis confirms it, an upstream node could give **minutes of warning** downstream, moving TabaPort from 'where is water now' to 'where will water be next'.
- **Recently rebuilt channels rank low for sensors.** The Nakivubo section at St Balikuddembe market stays on the map but is not a first site, in line with Eric's rule against sites under reconstruction.

Proposed first-sensor shortlist (subject to site screening and KCCA permission)

1. TP-04 Mulago roundabout (hospital access, top of the northern chain). **2. TP-03 Kalerwe** (named in every team document). **3. TP-01 Clock Tower** (headline blackspot; check KCCA works first). **4. TP-11 Nalukolongo** (channel reported undersized to Parliament, July 2026). **5. TP-02 Bwaise** (downstream end of the chain).

6. Flood Susceptibility Model (Google Earth Engine)

The model scores every 30 m cell of Kampala from 0 (dry ridge) to 1 (valley floor that floods first), tests the score against the register, and exports files for the team. It uses only open datasets in the Earth Engine catalogue, so it costs nothing to run and anyone on the team can repeat it.

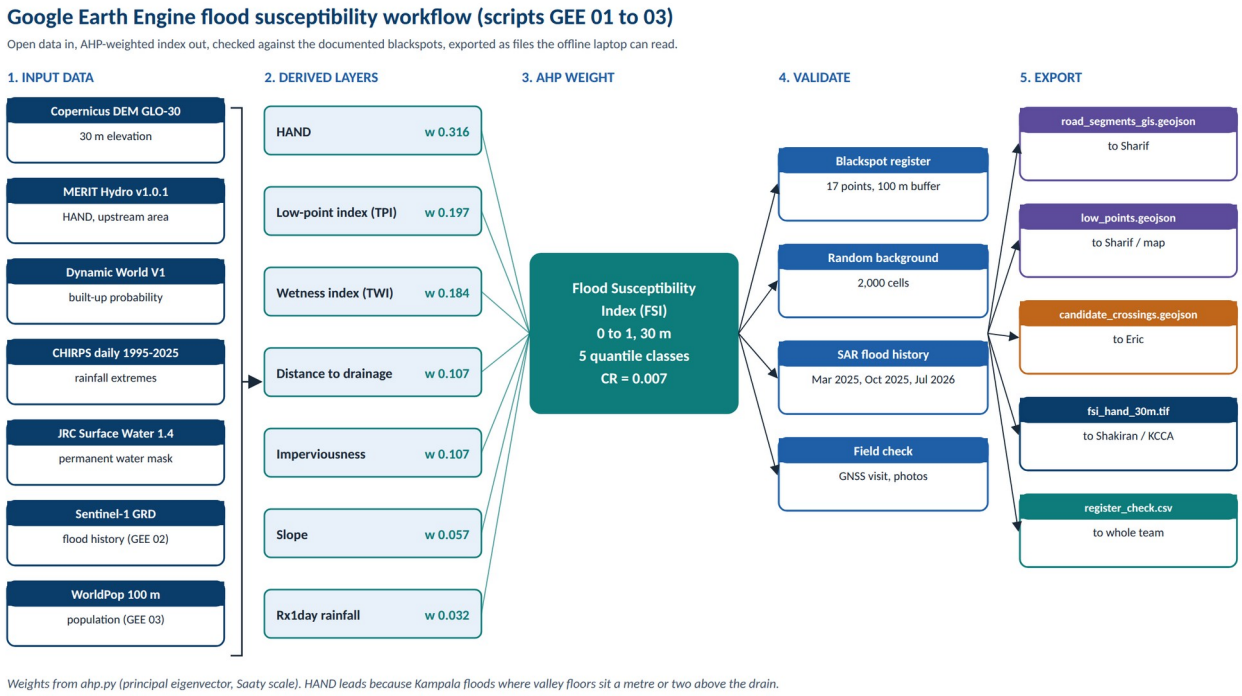


Figure 4. *Seven open inputs become seven layers, weighted by the Analytical Hierarchy Process into one Flood Susceptibility Index, validated against documented blackspots and radar history, then exported as files.*

Criterion	Data (Earth Engine)	Direction	Weight
Height above nearest drainage (HAND)	MERIT Hydro v1.0.1	Lower = more prone	0.316
Low-point index (local hollow)	Copernicus DEM GLO-30	Deeper hollow = more prone	0.197
Topographic wetness index	MERIT Hydro	Wetter = more prone	0.184
Distance to drainage	MERIT Hydro	Nearer = more prone	0.107
Imperviousness	Dynamic World V1	More sealed = more prone	0.107
Slope	Copernicus DEM GLO-30	Flatter = more prone	0.057
Rainfall extremes	CHIRPS Daily, 1995 to 2025	Wetter = more prone	0.032
Consistency ratio	Saaty eigenvector method	Limit 0.10	**0.007**

HAND and local hollows carry the most weight because, across a 15 km city, rainfall differs little while the height of a place above its drain differs a great deal. **Low points** are cells in the top class that sit under 5 m above drainage and at least 1 m below their surroundings; they are exported as polygons for the dispatcher map.

6.1 Satellite Flood History, and an Honest Limit

A second script compares Sentinel-1 radar before and after the March 2025, October 2025 and July 2026 events to map standing water. Radar passes every 6 to 12 days and struggles between tall buildings, so it **mostly misses Kampala's flash floods**, which drain within hours. That limit is itself the case for TabaPort: space-based monitoring cannot see the event this product is built for; a sensor in the culvert can.

6.2 Validation Test

The index is sampled within 100 m of each register point and compared with 2,000 random cells. **Acceptance test: at least 12 of the 17 register points fall in the top two classes.** If not, the weights are revisited with the team before any map is used in the video. The scripts are written and syntax-checked; results will be reported to MIIC once run.

7. Offline Map Data for the Dispatch Hub

The dispatcher laptop has no reliable network during a flood. The GIS component therefore does all spatial computation **before** the rain and delivers plain files to one folder. The laptop needs no GIS server, no cloud tiles and no API key, which keeps the software simple and dependable.

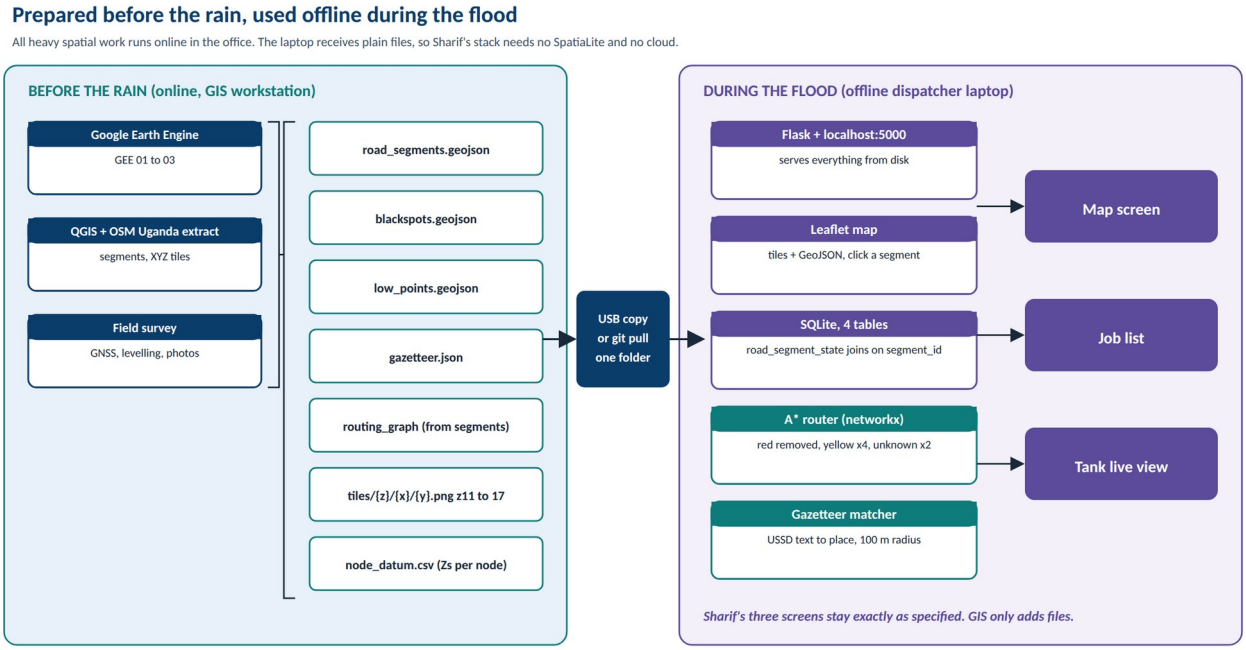


Figure 5. *Before the rain, Earth Engine, QGIS and field survey produce plain files. During the flood, the dispatcher screens run from disk, with two small GIS modules (router and gazetteer) behind the map.*

File	Contents	Refresh
road_segments.geojson	Kampala roads cut into 100 m segments with IDs and flood attributes	When roads or KCCA works change
blackspots.geojson	Register TP-01 to TP-17 with verified positions	After each field visit
low_points.geojson	Ponding areas from the susceptibility model	Once per season
gazetteer.json	Place names and local spellings for text reports	When a new spelling appears
tiles/	Base map rendered locally, zoom 11 to 17 (about 6,600 tiles)	Twice a year
node_datum.csv	Surveyed heights and thresholds for each sensor	After any re-mount

Licensing. The OpenStreetMap Foundation's tile policy states that offline use is not permitted on its public tile servers. TabaPort's base map is therefore rendered locally from OpenStreetMap data under the ODbL licence, with attribution. Coordinates follow WGS 84 for the web map and UTM zone 36N (EPSG:32636) for survey work.

8. Sensor Siting and Survey

Eric's technical note sets the rules for where a sensor may sit on a structure. The GIS component wraps those rules with a city-wide screen that finds candidate crossings, and a survey that makes every reading a real elevation.

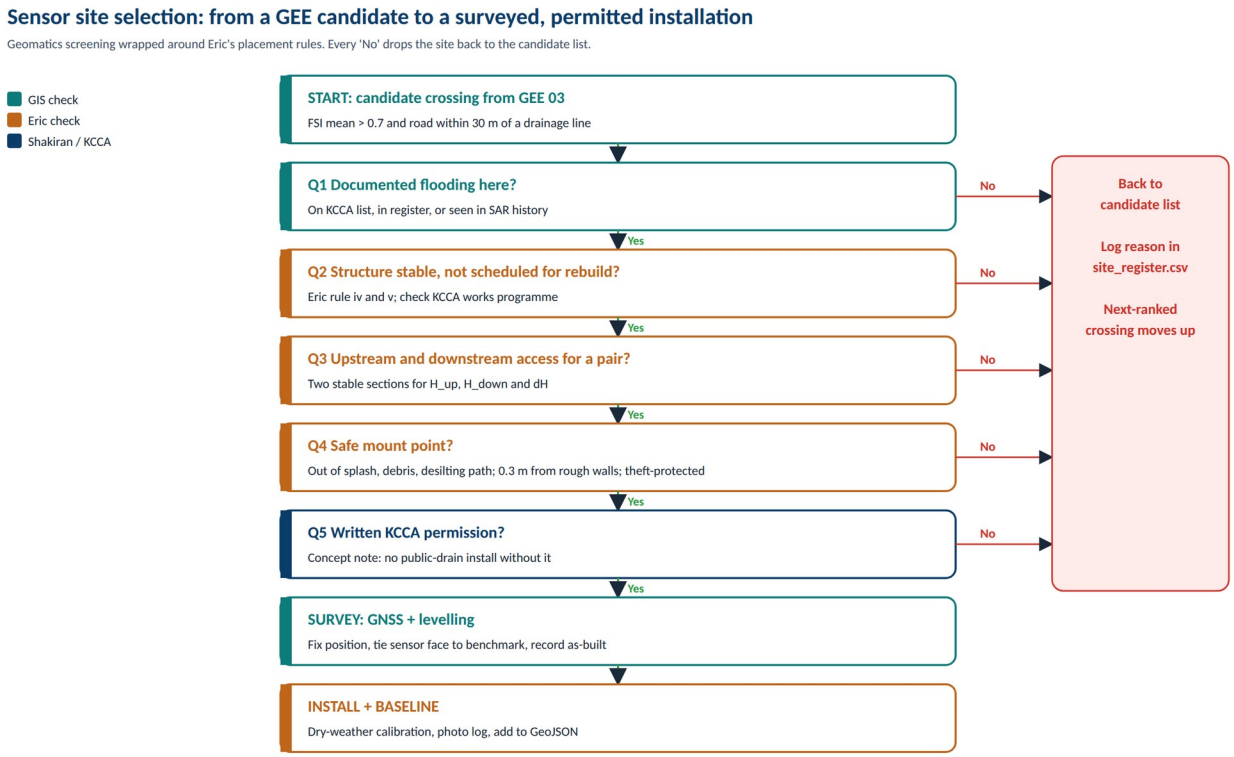


Figure 6. *Site selection flow. GIS screens and surveys; Eric decides structure and mounting; the project lead secures KCCA permission. Any 'No' returns the site to the list with the reason logged.*

A sensor reports an air gap in centimetres. Levelled from a survey benchmark, that becomes a **water surface elevation** comparable across every node and with the road itself. Two outcomes matter to KCCA:

- **Blockage detection:** the upstream minus downstream water elevation at a crossing (Eric's head difference) only works when both sensors share one surveyed datum.
- **Thresholds from measured geometry:** with the road surface surveyed, a segment turns **yellow** as water approaches the road and **red** when water reaches it, instead of relying on guessed numbers. Exact values will be set with Eric and the project lead during the trial.

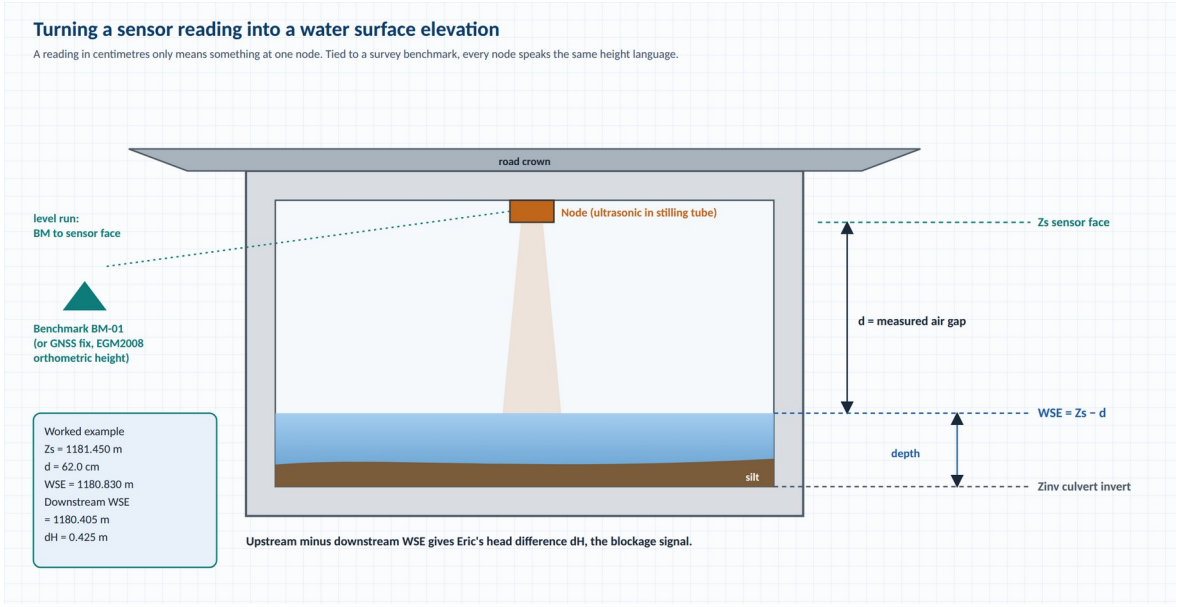


Figure 7. *Turning a reading into a water surface elevation. The worked example uses illustrative numbers.*

9. Dispatch Priority, Routing and Field Reports

9.1 Ranking Jobs

The IEEE entry ranks flood sites on three criteria. The GIS component pre-computes two of them for every road segment, so during a storm the hub only multiplies three numbers and sorts. Proposed weights, for the project lead to confirm:

Criterion	Source	Proposed weight
Velocity of water rise	Sensor reading	0.540
Proximity to critical infrastructure	GIS: distance to hospitals	0.297
Neighbourhood population density	GIS: WorldPop within 500 m	0.163
Consistency ratio	Saaty eigenvector method	**0.008**

9.2 Routing Crews Around Water

Each road segment costs its length times a factor: **green x1, unknown x2, yellow x4, red removed**. Treating a silent sensor as costlier than green honours the concept note's rule that silence is not safety. High ground is measured per segment, not assumed from a road's name.

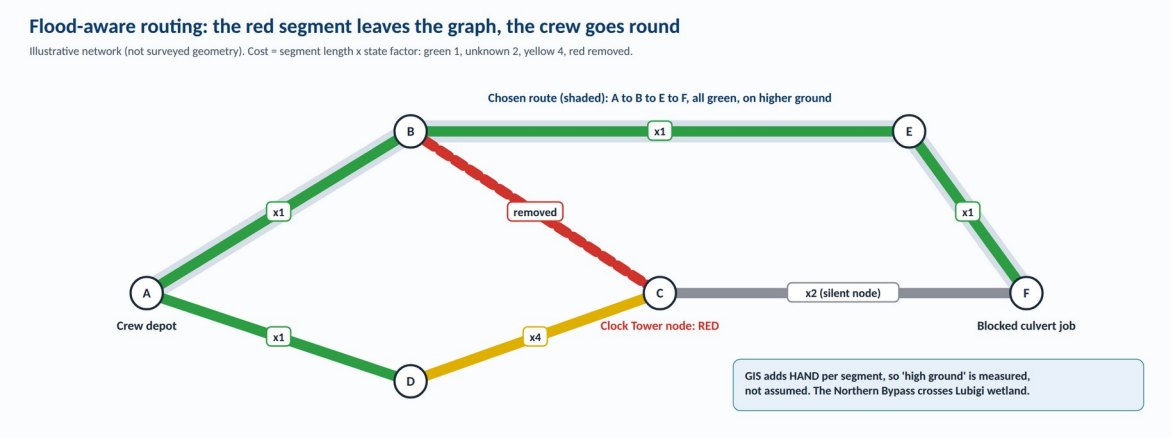


Figure 8. *Illustrative routing: the red segment leaves the network and the crew takes the all-green detour.*

9.3 Reading Boda Reports

A gazetteer of place names and the spellings people actually type (Bwayise, Natete, Kalerwe mkt, clocktower) turns a text report into a map location with a 100 m circle. Tested on sample messages, it matched all four real place mentions and correctly rejected an out-of-area message. It stores places, never people, alongside the team's phone-number hashing.

9.4 Is a Cluster Real?

After each storm, the Getis-Ord Gi* statistic (used in my LifeShield work) tests whether reports and red segments cluster more than chance allows. Over the 30-day trial this shows KCCA which culverts fail together and where the next sensor should go.

10. Shared System Model and Work Plan

The semantic model gives every object in TabaPort one name, one owner and a defined relationship to the others, so the four team documents describe one system.

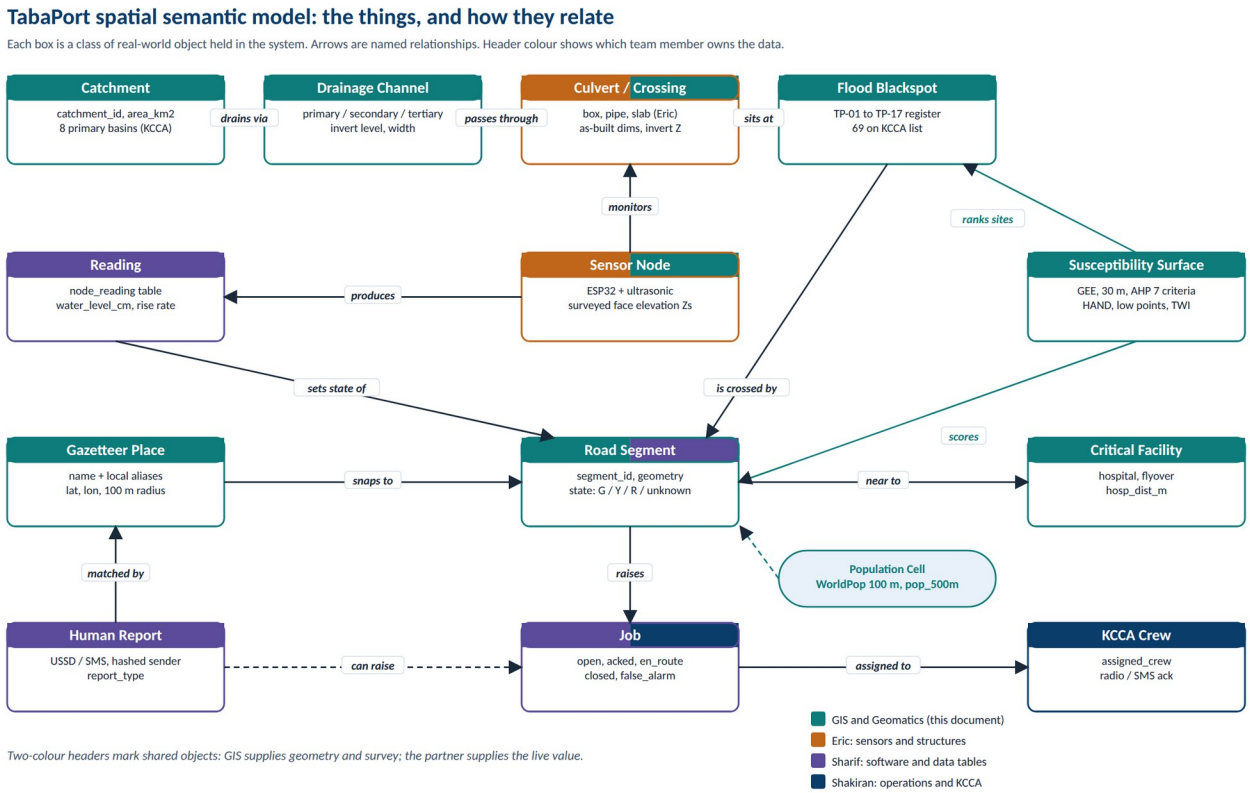


Figure 9. *TabaPort semantic model. Header colours show the owner; two-colour headers are shared objects where GIS supplies geometry and survey and the partner supplies the live value.*



Figure 10. *GIS work plan. Desk work supports the 9 October Phase 3 video; field geomatics follows only after KCCA gives written permission.*

Milestone	Measure of success	When
Susceptibility model run and validated	At least 12 of 17 register points in the top two classes	By 2 October 2026
Map files loaded in the dispatcher map	Segments, tiles and register open offline on localhost and a segment can be clicked end to end	By 6 October 2026
Map scenes for the Phase 3 video	Delivered to the project lead	By 9 October 2026
Field verification of top 5 sites	GNSS position and photos for each; register updated	Within 2 weeks of KCCA permission
Datum survey of the trial node pair	Sensor, invert and road heights levelled from a benchmark; loop closed	Before trial start

Milestone	Measure of success	When
30-day trial support	Every job located to a segment; weekly map check; Gi* summary at the end	Trial period

11. What the GIS Component Asks of the TES Programme

These requests sit inside the concept note's overall ask for lab access, mentorship and light catalytic support to finish Generation 1 and prepare a 30-day trial.

Request	Why it matters
Mentorship from a hydrologist or drainage engineer	Independent review of the susceptibility weights and the yellow / red thresholds before they guide a real crew
Introduction to KCCA's drainage team	Access to the official 69-blackspot list and as-built drawings, and the written permission every field site needs
Access to survey-grade GNSS and a level for five sites	Centimetre-level heights are what make blockage detection and physical thresholds possible
Light field support (transport to sites, printing of site packs)	Five site visits and a KCCA permission map pack

12. Honest Limits

- **A 30 m elevation model cannot see a kerb, a culvert or a one-metre road dip.** The index points the survey; it is not a hydraulic model.
- **Hydrology data is about 90 m**, so heights above small tertiary drains are approximate until checked on the ground.
- **Radar misses most flash floods**, so the satellite history under-counts events.
- **13 of 17 register coordinates are estimates** and will not be used for dispatch until GNSS-verified.
- **OpenStreetMap is incomplete in dense areas** such as Bwaise; YouthMappers mapathons and field checks will fill the gaps.
- **KCCA drainage works change geometry**; rebuilt sections are re-surveyed before their data is trusted.
- **No field installation or structure survey without written KCCA permission.**

Closing statement

If the map cannot say where the water is going, the dispatcher is guessing. The GIS component makes sure TabaPort knows Kampala's ground **before** the rain arrives, so that when the network fails, the system still knows where the city floods and how to reach it safely.

Appendix A. Google Earth Engine Scripts

Included to show the work is implementable. Asset paths under projects/tabaport/ are placeholders for the team's Earth Engine project.

A.1 Flood susceptibility, low points and blackspot check

```
/**************************************************************************
 * TabaPort GEE 01: Flood susceptibility, low points and blackspot check
 * Kampala, Uganda. IEEE Response Quest Concept #5267.
 * Workstream: GIS and Geomatics.
 * Output CRS for all exports: WGS 84 / UTM zone 36N (EPSG:32636).
 **************************************************************************/

// 1. AREA OF INTEREST -----------------------------------------------------
// Replace with the KCCA boundary asset when uploaded:
// var aoi = ee.FeatureCollection('projects/tabaport/assets/kcca_boundary').geometry();
var aoi = ee.Geometry.Rectangle([32.49, 0.22, 32.68, 0.41]);
Map.centerObject(aoi, 12);
Map.setOptions('HYBRID');

// 2. FLOOD LOCATION REGISTER (indicative points, to be replaced by GNSS fixes)
var register = ee.FeatureCollection([
  ee.Feature(ee.Geometry.Point([32.5785, 0.3095]), {id: 'TP-01', name: 'Clock Tower'}),
  ee.Feature(ee.Geometry.Point([32.5611, 0.3500]), {id: 'TP-02', name: 'Bwaise'}),
  ee.Feature(ee.Geometry.Point([32.5720, 0.3525]), {id: 'TP-03', name: 'Kalerwe'}),
  ee.Feature(ee.Geometry.Point([32.5745, 0.3395]), {id: 'TP-04', name: 'Mulago roundabout'}),
  ee.Feature(ee.Geometry.Point([32.5690, 0.3460]), {id: 'TP-05', name: 'Kubiri roundabout'}),
  ee.Feature(ee.Geometry.Point([32.5500, 0.3470]), {id: 'TP-06', name: 'Northern Bypass Lubigi'}),
  ee.Feature(ee.Geometry.Point([32.5775, 0.3110]), {id: 'TP-07', name: 'Nakivubo St Balikuddembe'}),
  ee.Feature(ee.Geometry.Point([32.5990, 0.3285]), {id: 'TP-08', name: 'Lugogo Bypass'}),
  ee.Feature(ee.Geometry.Point([32.5930, 0.3070]), {id: 'TP-09', name: 'Mukwano Rd'}),
  ee.Feature(ee.Geometry.Point([32.5350, 0.2983]), {id: 'TP-10', name: 'Nateete'}),
  ee.Feature(ee.Geometry.Point([32.5490, 0.2960]), {id: 'TP-11', name: 'Nalukolongo'}),
  ee.Feature(ee.Geometry.Point([32.5610, 0.2940]), {id: 'TP-12', name: 'Ndeeba'}),
  ee.Feature(ee.Geometry.Point([32.6330, 0.3350]), {id: 'TP-13', name: 'Kinawataka'}),
  ee.Feature(ee.Geometry.Point([32.6290, 0.3480]), {id: 'TP-14', name: 'Kyambogo'}),
  ee.Feature(ee.Geometry.Point([32.6144, 0.3550]), {id: 'TP-15', name: 'Ntinda'}),
  ee.Feature(ee.Geometry.Point([32.6078, 0.2872]), {id: 'TP-16', name: 'Kansanga'}),
  ee.Feature(ee.Geometry.Point([32.6530, 0.2890]), {id: 'TP-17', name: 'Port Bell'})
]);

// 3. TERRAIN AND HYDROLOGY ------------------------------------------------
var dem = ee.ImageCollection('COPERNICUS/DEM/GLO30')
  .filterBounds(aoi).select('DEM').mosaic()
  .setDefaultProjection({crs: 'EPSG:4326', scale: 30})
  .rename('elev').clip(aoi);

var slope = ee.Terrain.slope(dem).rename('slope');                // degrees
var merit = ee.Image('MERIT/Hydro/v1_0_1').clip(aoi);
var hand  = merit.select('hnd').rename('hand');                     // m above nearest drainage
var upa   = merit.select('upa');                                    // upstream area, km2

// Topographic Wetness Index: ln(a / tan b)
var sca = upa.multiply(1e6).divide(90);                             // m2 per m contour width
var slopeRad = ee.Terrain.slope(merit.select('elv'))
  .multiply(Math.PI / 180).max(0.001);
var twi = sca.log().subtract(slopeRad.tan().log()).rename('twi');

// Low-point index: elevation relative to the 300 m neighbourhood mean.
// Negative values = hollows and road dips where water ponds.
var localMean = dem.reduceNeighborhood({
  reducer: ee.Reducer.mean(), kernel: ee.Kernel.circle(300, 'meters')});
var tpi = dem.subtract(localMean).rename('tpi');

// Distance to drainage lines (cells draining at least 1 km2).
var streams = upa.gt(1).selfMask();
var distDrain = streams.fastDistanceTransform(100).sqrt()
  .multiply(90).rename('dist_drain')
  .reproject({crs: 'EPSG:32636', scale: 90});

// 4. SURFACE AND RAINFALL -------------------------------------------------
var built = ee.ImageCollection('GOOGLE/DYNAMICWORLD/V1')
  .filterBounds(aoi).filterDate('2025-01-01', '2026-01-01')
  .select('built').mean().rename('built').clip(aoi);                // 0 to 1

var years = ee.List.sequence(1995, 2025);
var rx1day = ee.ImageCollection(years.map(function (y) {
  y = ee.Number(y);
  return ee.ImageCollection('UCSB-CHG/CHIRPS/DAILY')
    .filter(ee.Filter.calendarRange(y, y, 'year'))
    .select('precipitation').max();
})).mean().rename('rx1day').clip(aoi);                              // mean annual max 1-day rain, mm

var permanentWater = ee.Image('JRC/GSW1_4/GlobalSurfaceWater')
  .select('occurrence').gt(80).unmask(0).clip(aoi);
```

```
// 5. NORMALISE (2nd to 98th percentile) ------------------------------------
function norm(img, invert) {
  var band = ee.String(img.bandNames().get(0));
  var p = img.reduceRegion({reducer: ee.Reducer.percentile([2, 98]),
    geometry: aoi, scale: 90, maxPixels: 1e10, bestEffort: true});
  var lo = ee.Number(p.get(band.cat('_p2')));
  var hi = ee.Number(p.get(band.cat('_p98')));
  var s = img.subtract(lo).divide(hi.subtract(lo)).clamp(0, 1);
  return invert ? ee.Image(1).subtract(s) : s;
}
var nHand  = norm(hand, true);       // low HAND  = high susceptibility
var nTpi   = norm(tpi, true);        // hollow    = high
var nTwi   = norm(twi, false);       // wet       = high
var nDist  = norm(distDrain, true);  // near drain = high
var nBuilt = norm(built, false);     // sealed    = high
var nSlope = norm(slope, true);      // flat      = high
var nRain  = norm(rx1day, false);    // wetter    = high

// 6. AHP WEIGHTS (CR = 0.007, computed in ahp.py) -------------------------
var fsi = nHand.multiply(0.316)
  .add(nTpi.multiply(0.197))
  .add(nTwi.multiply(0.184))
  .add(nDist.multiply(0.107))
  .add(nBuilt.multiply(0.107))
  .add(nSlope.multiply(0.057))
  .add(nRain.multiply(0.032))
  .where(permanentWater.eq(1), 1)
  .rename('fsi').clip(aoi);

// Five classes by quantile: 1 very low ... 5 very high
var q = fsi.reduceRegion({reducer: ee.Reducer.percentile([20, 40, 60, 80]),
  geometry: aoi, scale: 30, maxPixels: 1e10, bestEffort: true});
var fsiClass = ee.Image(1)
  .where(fsi.gt(ee.Number(q.get('fsi_p20'))), 2)
  .where(fsi.gt(ee.Number(q.get('fsi_p40'))), 3)
  .where(fsi.gt(ee.Number(q.get('fsi_p60'))), 4)
  .where(fsi.gt(ee.Number(q.get('fsi_p80'))), 5)
  .rename('fsi_class').clip(aoi);

// Low points: very high class AND within 5 m of drainage AND a local hollow
var lowPoints = fsiClass.eq(5).and(hand.lt(5)).and(tpi.lt(-1)).selfMask();

// 7. MAP ------------------------------------------------------------------
var pal = ['1a9850', '91cf60', 'fee08b', 'fc8d59', 'd73027'];
Map.addLayer(dem, {min: 1130, max: 1310, palette: ['08306b', 'f7fbff']}, 'Elevation (m)', false);
Map.addLayer(hand, {min: 0, max: 30, palette: ['08519c', 'deebf7']}, 'HAND (m)', false);
Map.addLayer(fsiClass, {min: 1, max: 5, palette: pal}, 'Flood susceptibility class');
Map.addLayer(lowPoints, {palette: ['7a0177']}, 'Low points (ponding)');
Map.addLayer(streams, {palette: ['00bfff']}, 'Drainage lines (>1 km2)', false);
Map.addLayer(register, {color: 'black'}, 'Flood location register');

// 8. VALIDATION: do documented blackspots fall in high classes? ------------
var atRegister = fsi.addBands(fsiClass).addBands(hand).addBands(tpi)
  .reduceRegions({collection: register.map(function (f) {
    return f.buffer(100); }),                                  // 100 m, same as IEEE text-report radius
  reducer: ee.Reducer.max(), scale: 30});
print('Register sampled (max within 100 m):', atRegister);

var background = fsi.sample({region: aoi, scale: 30, numPixels: 2000, seed: 42, geometries: false});
print('Mean FSI at register points:', atRegister.aggregate_mean('fsi'));
print('Mean FSI at 2000 random cells:', background.aggregate_mean('fsi'));
print('Share of register in class 4 or 5:',
  atRegister.filter(ee.Filter.gte('fsi_class', 4)).size()
    .divide(atRegister.size()));

// 9. EXPORTS (feed Sharif's offline Leaflet stack) ------------------------
Export.image.toDrive({image: fsi.addBands(fsiClass).addBands(hand).float(),
  description: 'tabaport_fsi_hand_30m', folder: 'TabaPort_GIS',
  region: aoi, scale: 30, crs: 'EPSG:32636', maxPixels: 1e10});

var lowPolys = lowPoints.reduceToVectors({geometry: aoi, scale: 30,
  geometryType: 'polygon', eightConnected: true, maxPixels: 1e10})
  .map(function (f) { return f.set('area_m2', f.geometry().area(1)); })
  .filter(ee.Filter.gt('area_m2', 2000));                      // drop specks under 0.2 ha
Export.table.toDrive({collection: lowPolys, description: 'tabaport_low_points',
  folder: 'TabaPort_GIS', fileFormat: 'GeoJSON'});

Export.table.toDrive({collection: atRegister, description: 'tabaport_register_check',
  folder: 'TabaPort_GIS', fileFormat: 'CSV'});
```

A.2 Sentinel-1 flood history

```
/**********************************************************************
 * TabaPort GEE 02: Sentinel-1 SAR flood history for recorded Kampala events
 * Change detection: post-event minimum vs pre-event median backscatter (VV).
 * Honest limit: a 6 to 12 day revisit and urban radar shadow mean flash
 * floods that drain within hours are often missed. What SAR does catch is
 * slow-draining valley water, which is exactly where culvert nodes belong.
```

```
 ********************************************************************************/
var aoi = ee.Geometry.Rectangle([32.49, 0.22, 32.68, 0.41]);
Map.centerObject(aoi, 12);

// Recorded events (edit dates from the KCCA incident log when available)
var events = [
  {name: 'Mar2025', pre: ['2025-01-15', '2025-03-24'], post: ['2025-03-26', '2025-04-05']},  // GDACS event
1103187
  {name: 'Oct2025', pre: ['2025-08-15', '2025-10-29'], post: ['2025-10-31', '2025-11-10']},  // Nakivubo
flooding 31 Oct 2025
  {name: 'Jul2026', pre: ['2026-05-15', '2026-07-01'], post: ['2026-07-02', '2026-07-31']}   // Nalukolongo,
Lugogo, Kyambogo
];

var hand  = ee.Image('MERIT/Hydro/v1_0_1').select('hnd');
var slope = ee.Terrain.slope(ee.ImageCollection('COPERNICUS/DEM/GLO30')
  .filterBounds(aoi).select('DEM').mosaic()
  .setDefaultProjection({crs: 'EPSG:4326', scale: 30}));
var permWater = ee.Image('JRC/GSW1_4/GlobalSurfaceWater').select('occurrence').gt(80).unmask(0);

function s1(start, end) {
  return ee.ImageCollection('COPERNICUS/S1_GRD')
    .filterBounds(aoi).filterDate(start, end)
    .filter(ee.Filter.eq('instrumentMode', 'IW'))
    .filter(ee.Filter.listContains('transmitterReceiverPolarisation', 'VV'))
    .select('VV')
    .map(function (i) { return i.focalMedian(50, 'circle', 'meters'); }); // speckle
}

var floodMaps = events.map(function (e) {
  var preCol = s1(e.pre[0], e.pre[1]);
  var postCol = s1(e.post[0], e.post[1]);
  print(e.name + ' scenes (pre, post):', preCol.size(), postCol.size());
  var pre = preCol.median();
  var post = postCol.min();
  var flooded = post.subtract(pre).lt(-3)       // backscatter drop of 3 dB or more
    .and(post.lt(-15))                          // smooth open water
    .and(hand.lt(10))                           // physically plausible: near drainage
    .and(slope.lt(5))
    .and(permWater.eq(0))                       // not the lake
    .rename('flood').unmask(0).clip(aoi);
  Map.addLayer(flooded.selfMask(), {palette: ['0000ff']}, 'SAR flood ' + e.name, false);
  return flooded.set('event', e.name);
});

// Flood frequency: how many recorded events flooded each cell
var freq = ee.ImageCollection(floodMaps).sum().rename('flood_count').clip(aoi);
Map.addLayer(freq.selfMask(), {min: 1, max: events.length, palette: ['c6dbef', '2171b5', '08306b']},
  'Flood frequency (events)');

Export.image.toDrive({image: freq.toByte(), description: 'tabaport_sar_flood_frequency',
  folder: 'TabaPort_GIS', region: aoi, scale: 20, crs: 'EPSG:32636', maxPixels: 1e10});
```

A.3 Segment attributes and candidate sensor crossings

```
/**********************************************************************
 * TabaPort GEE 03: Road segment attributes, AHP exposure inputs and
 * candidate sensor crossings. Output GeoJSON loads straight into
 * Sharif's Leaflet layer and joins road_segment_state on segment_id.
 * Inputs uploaded from QGIS (OSM Uganda extract, split at 100 m):
 *   kampala_road_segments  (LineString, property segment_id e.g. clock_tower_01)
 *   kampala_hospitals      (Point, from OSM amenity=hospital, checked in field)
 **********************************************************************/
var aoi = ee.Geometry.Rectangle([32.49, 0.22, 32.68, 0.41]);
var segments  = ee.FeatureCollection('projects/tabaport/assets/kampala_road_segments');
var hospitals = ee.FeatureCollection('projects/tabaport/assets/kampala_hospitals');
var fsiStack  = ee.Image('projects/tabaport/assets/tabaport_fsi_hand_30m'); // export of GEE 01

// Distance to nearest hospital (m), searched to 5 km
var hospDist = hospitals.distance({searchRadius: 5000, maxError: 10}).rename('hosp_dist_m');

// People within 500 m (WorldPop 100 m, 2020)
var pop = ee.ImageCollection('WorldPop/GP/100m/pop')
  .filter(ee.Filter.eq('country', 'UGA')).filter(ee.Filter.eq('year', 2020)).first();
var pop500 = pop.reduceNeighborhood({reducer: ee.Reducer.sum(),
  kernel: ee.Kernel.circle(500, 'meters')}).rename('pop_500m');

// Drainage distance again for crossing detection
var upa = ee.Image('MERIT/Hydro/v1_0_1').select('upa');
var distDrain = upa.gt(1).selfMask().fastDistanceTransform(100).sqrt()
  .multiply(90).rename('dist_drain').reproject({crs: 'EPSG:32636', scale: 90});

var stack = fsiStack.select(['fsi', 'hand'])
  .addBands(hospDist).addBands(pop500).addBands(distDrain);

var attrs = stack.reduceRegions({
  collection: segments,
  reducer: ee.Reducer.mean().combine({reducer2: ee.Reducer.min(), sharedInputs: true}),
```

```
  scale: 30
}).map(function (f) {
  // Static AHP exposure inputs, scaled 0 to 1 for the dispatch engine
  var infra = ee.Number(1).subtract(ee.Number(f.get('hosp_dist_m_min')).divide(5000)).max(0);
  var popn  = ee.Number(f.get('pop_500m_mean')).divide(40000).min(1);
  return f.set({
    infra_score: infra, pop_score: popn,
    low_point: ee.Number(f.get('hand_min')).lt(3),
    static_class: ee.Algorithms.If(ee.Number(f.get('fsi_mean')).gt(0.7), 'high',
                  ee.Algorithms.If(ee.Number(f.get('fsi_mean')).gt(0.5), 'medium', 'low'))
  });
});

// Candidate sensor crossings: high susceptibility AND the road crosses a drainage line
var candidates = attrs
  .filter(ee.Filter.gt('fsi_mean', 0.7))
  .filter(ee.Filter.lt('dist_drain_min', 30))
  .sort('fsi_mean', false);
print('Candidate crossings for Eric (top 25):', candidates.limit(25));

Map.addLayer(attrs, {color: '555555'}, 'Segments');
Map.addLayer(candidates, {color: 'd73027'}, 'Candidate sensor crossings');

Export.table.toDrive({collection: attrs, description: 'road_segments_gis',
  folder: 'TabaPort_GIS', fileFormat: 'GeoJSON'});
Export.table.toDrive({collection: candidates, description: 'candidate_sensor_crossings',
  folder: 'TabaPort_GIS', fileFormat: 'GeoJSON'});
```

Appendix B. Offline Python Toolkit

Runs on the dispatcher laptop with Python 3, numpy and networkx; results quoted in this report come from running it.

B.1 ahp.py

```
"""TabaPort GIS: Analytical Hierarchy Process weights and consistency ratio.
Author: TabaPort GIS and Geomatics workstream.
Runs offline with numpy only."""
import numpy as np

RI = {1: 0.0, 2: 0.0, 3: 0.58, 4: 0.90, 5: 1.12, 6: 1.24, 7: 1.32, 8: 1.41, 9: 1.45}

def ahp(names, upper):
    """upper: dict {(i, j): value} for i < j on the Saaty 1 to 9 scale."""
    n = len(names)
    A = np.ones((n, n))
    for (i, j), v in upper.items():
        A[i, j] = v
        A[j, i] = 1.0 / v
    vals, vecs = np.linalg.eig(A)
    k = np.argmax(vals.real)
    w = np.abs(vecs[:, k].real)
    w = w / w.sum()
    lam = vals[k].real
    ci = (lam - n) / (n - 1)
    cr = ci / RI[n] if RI[n] else 0.0
    return A, w, lam, ci, cr

# 1. Static flood susceptibility (computed in Google Earth Engine)
S_NAMES = ["HAND", "Low-point index", "Flow accumulation (TWI)",
           "Distance to drainage", "Imperviousness", "Slope", "Rainfall extremes"]
S_UPPER = {
    (0, 1): 2, (0, 2): 2, (0, 3): 3, (0, 4): 3, (0, 5): 5, (0, 6): 7,
    (1, 2): 1, (1, 3): 2, (1, 4): 2, (1, 5): 4, (1, 6): 6,
    (2, 3): 2, (2, 4): 2, (2, 5): 3, (2, 6): 5,
    (3, 4): 1, (3, 5): 2, (3, 6): 4,
    (4, 5): 2, (4, 6): 4,
    (5, 6): 2,
}

# 2. Dispatch Tactical Priority Score (the team's three IEEE criteria)
D_NAMES = ["Velocity of water rise", "Proximity to critical infrastructure",
           "Neighbourhood population density"]
D_UPPER = {(0, 1): 2, (0, 2): 3, (1, 2): 2}

if __name__ == "__main__":
    for title, names, up in [("STATIC FLOOD SUSCEPTIBILITY", S_NAMES, S_UPPER),
                             ("DISPATCH TACTICAL PRIORITY", D_NAMES, D_UPPER)]:
        A, w, lam, ci, cr = ahp(names, up)
        print(title)
        for nm, wi in zip(names, w):
            print(f"  {nm:40s} {wi:.3f}")
        print(f"  lambda_max={lam:.3f}  CI={ci:.4f}  CR={cr:.4f}  "
              f"{'CONSISTENT (CR < 0.10)' if cr < 0.1 else 'REVISE'}")
```

```
        print()
```

B.2 tabaport_gis_offline.py

```
"""TabaPort GIS offline toolkit (runs on the dispatcher laptop, no internet).
1. Gazetteer matching for boda USSD/SMS reports (IEEE #5267: 'regex tokenizer + gazetteer').
2. Stage elevation and head difference from sensor readings (Eric's H_up, H_down, dH).
3. Flood-aware A* routing over pre-computed GeoJSON road segments.
Dependencies: Python 3 standard library + networkx."""
import json, math, re, difflib
import networkx as nx

# ---------- 1. Gazetteer ----------------------------------------------------
GAZETTEER = [
    {"place_id": "TP-01", "name": "Clock Tower", "aliases": ["clocktower", "clock tower junction",
"queensway"], "lat": 0.3095, "lon": 32.5785, "segment_ids": ["clock_tower_01", "clock_tower_02"]},
    {"place_id": "TP-02", "name": "Bwaise", "aliases": ["bwayise", "bwaise market", "bwaise roundabout"],
"lat": 0.3500, "lon": 32.5611, "segment_ids": ["bwaise_01"]},
    {"place_id": "TP-03", "name": "Kalerwe", "aliases": ["kalerwe market", "kalerwe mkt", "kaleerwe"], "lat":
0.3525, "lon": 32.5720, "segment_ids": ["kalerwe_01"]},
    {"place_id": "TP-04", "name": "Mulago roundabout", "aliases": ["mulago", "mulago round about"], "lat":
0.3395, "lon": 32.5745, "segment_ids": ["mulago_rbt_01"]},
    {"place_id": "TP-11", "name": "Nalukolongo", "aliases": ["nalukolongo channel", "nalukolongo bridge"],
"lat": 0.2960, "lon": 32.5490, "segment_ids": ["nalukolongo_01"]},
    {"place_id": "TP-10", "name": "Nateete", "aliases": ["natete", "family loaf"], "lat": 0.2983, "lon":
32.5350, "segment_ids": ["nateete_01"]},
]
UNCERTAINTY_M = 100  # IEEE #5267: text reports carry a 100 m radius until confirmed

def _norm(t):
    return re.sub(r"[^a-z ]", " ", t.lower()).split()

def match_report(text, cutoff=0.82):
    """Return best gazetteer place for free text, or None. Uses 1 to 3 word n-grams."""
    words = _norm(text)
    grams = {" ".join(words[i:i + n]) for n in (1, 2, 3) for i in range(len(words) - n + 1)}
    best, best_score = None, 0.0
    for p in GAZETTEER:
        for label in [p["name"].lower()] + p["aliases"]:
            for g in grams:
                s = difflib.SequenceMatcher(None, g, label).ratio()
                if s > best_score:
                    best, best_score = p, s
    if best_score < cutoff:
        return None
    return {"place_id": best["place_id"], "name": best["name"], "score": round(best_score, 2),
            "lat": best["lat"], "lon": best["lon"], "radius_m": UNCERTAINTY_M,
            "candidate_segments": best["segment_ids"]}

# ---------- 2. Stage elevation and head difference -------------------------
def stage_elevation(sensor_face_elev_m, measured_distance_cm):
    """Water surface elevation (m above datum) = surveyed sensor face elevation - measured air gap."""
    return round(sensor_face_elev_m - measured_distance_cm / 100.0, 3)

def head_difference(wse_upstream_m, wse_downstream_m):
    """Eric's dH: a growing positive value at a crossing suggests blockage or restriction."""
    return round(wse_upstream_m - wse_downstream_m, 3)

# ---------- 3. Flood-aware routing -----------------------------------------
STATE_FACTOR = {"green": 1.0, "unknown": 2.0, "yellow": 4.0, "red": math.inf}

def haversine_m(a, b):
    R = 6371000.0
    la1, lo1, la2, lo2 = map(math.radians, (a[1], a[0], b[1], b[0]))
    h = math.sin((la2 - la1) / 2) ** 2 + math.cos(la1) * math.cos(la2) * math.sin((lo2 - lo1) / 2) ** 2
    return 2 * R * math.asin(math.sqrt(h))

def build_graph(segments_geojson, states):
    """segments_geojson: FeatureCollection of LineStrings with properties.segment_id.
    states: {segment_id: 'green'|'yellow'|'red'|'unknown'} from SQLite road_segment_state."""
    G = nx.Graph()
    for f in segments_geojson["features"]:
        sid = f["properties"]["segment_id"]
        coords = f["geometry"]["coordinates"]
        u, v = tuple(coords[0]), tuple(coords[-1])
        length = sum(haversine_m(coords[i], coords[i + 1]) for i in range(len(coords) - 1))
        factor = STATE_FACTOR[states.get(sid, "unknown")]
        if math.isinf(factor):
            continue  # red segment: infinite penalty, removed from the graph
        G.add_edge(u, v, segment_id=sid, length_m=length, cost=length * factor)
    return G

def safe_route(G, start, goal):
    path = nx.astar_path(G, start, goal, heuristic=haversine_m, weight="cost")
    segs = [G.edges[path[i], path[i + 1]]["segment_id"] for i in range(len(path) - 1)]
    dist = sum(G.edges[path[i], path[i + 1]]["length_m"] for i in range(len(path) - 1))
    return segs, round(dist)
```

```
if __name__ == "__main__":
    print("GAZETTEER TESTS")
    for msg in ["Flooding at kalerwe mkt road closed", "BWAYISE water too much",
                "clocktower submerged", "natete near family loaf blocked drain", "all fine in Entebbe"]:
        print(f"  '{msg}' -> {match_report(msg)}")
    print("\nSTAGE AND HEAD DIFFERENCE TEST")
    up = stage_elevation(1181.450, 62.0)
    down = stage_elevation(1181.120, 71.5)
    print(f"  WSE up={up} m, WSE down={down} m, dH={head_difference(up, down)} m")
    print("\nROUTING TEST (demo network, not surveyed geometry)")
    A, B, C, D = (32.570, 0.340), (32.575, 0.345), (32.572, 0.350), (32.566, 0.347)
    demo = {"type": "FeatureCollection", "features": [
        {"type": "Feature", "properties": {"segment_id": "demo_direct"}, "geometry": {"type": "LineString",
"coordinates": [A, C]}},
        {"type": "Feature", "properties": {"segment_id": "demo_east_1"}, "geometry": {"type": "LineString",
"coordinates": [A, B]}},
        {"type": "Feature", "properties": {"segment_id": "demo_east_2"}, "geometry": {"type": "LineString",
"coordinates": [B, C]}},
        {"type": "Feature", "properties": {"segment_id": "demo_west_1"}, "geometry": {"type": "LineString",
"coordinates": [A, D]}},
        {"type": "Feature", "properties": {"segment_id": "demo_west_2"}, "geometry": {"type": "LineString",
"coordinates": [D, C]}}]}
    for states in [{}, {"demo_direct": "green"}, {"demo_direct": "red", "demo_east_1": "yellow",
                   "demo_east_2": "green", "demo_west_1": "green", "demo_west_2": "green"}]:
        G = build_graph(demo, states)
        print(f"  states={states or 'all unknown'} -> {safe_route(G, A, C)}")
```

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